ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOG

MODULE 4,PRACTICAL 3

UNIT:DATABASE

UNIT CODE: 0612 451 08A

Time allowed: 3 hours

Materials and Tools Required (Essential Items Only)

No.	Item	Specification	Qty per candidate
1	Computer	Working, with Microsoft Access installed and a screenshot facility (Snipping Tool or Print Screen)	1
2	Flash disk	For saving the database file	1
3	Answer sheet and pen	For SQL commands, screenshots and written answers	1 set

Instructions to the Candidate

1. Work individually and read all tasks before you begin.
2. Complete the tasks in the order given.
3. Type all SQL commands in the SQL view of a query and save each query with the name given in the task.
4. Record your observations and take screenshots of your results in your answer sheet.
5. Raise your hand when you see **"Call the assessor"**.
6. Save your database as `YourName_StudentDB.accdb`.

Table Structure

Table name: `Students`

Field Name	Data Type	Size / Notes
StudentID	Number (Integer)	Primary key
FirstName	Short Text	30 characters
LastName	Short Text	30 characters

Gender	Short Text	10 characters
Course	Short Text	30 characters
Age	Number (Integer)	
Fees	Currency	

Sample Data

Records to be entered using the form (Task 3):

StudentID	FirstName	LastName	Gender	Course	Age	Fees
101	Mary	Achieng	Female	ICT	19	25000
102	John	Otieno	Male	Electrical	21	28000
103	Faith	Wanjiru	Female	ICT	20	25000
104	Brian	Kamau	Male	Plumbing	22	22000
105	Grace	Atieno	Female	Electrical	18	28000

Record to be entered using the INSERT command (Task 6):

StudentID	FirstName	LastName	Gender	Course	Age	Fees
106	Peter	Odhiambo	Male	ICT	23	25000

Task 1: Create a New Database

Instructions:

1. Open Microsoft Access and create a **Blank Database**.
2. Name the database `YourName_StudentDB` and save it on your flash disk.
3. Take a screenshot showing the database name in the title bar.

Task 2: Create the Table Using the CREATE Command

Instructions:

1. Click **Create → Query Design**. Close the *Show Table* window without adding any table.
2. Switch to **SQL View**.
3. Type a `CREATE TABLE` command that creates the table **Students** with the fields, data types and sizes shown in the Table Structure. Make **StudentID** the primary key.
4. Click **Run**.
5. Save the query as `qryCreateTable`.
6. Open the table in **Design View** and take a screenshot showing the fields, data types and primary key.

Call the assessor to check the table structure.

Task 3: Insert Data Using a Form

Instructions:

1. Select the **Students** table, then click **Create → Form** to create a form.
2. Save the form as `frmStudents`.
3. Switch to **Form View** and enter the **five records** from the first sample data table. Use the form only. Do not type directly into the table.
4. Open the **Students** table and confirm that all five records are saved. Take a screenshot.

Task 4: Add Buttons to the Form

Instructions:

1. Open `frmStudents` in **Design View**.
2. Using the **Button** control (with the Command Button Wizard turned on), add the following buttons to the form footer or an empty area:
    - (a) A button that moves to the **Next Record**, with the caption **Next**.
    - (b) A button that moves to the **Previous Record**, with the caption **Previous**.
    - (c) A button that **adds a new record**, with the caption **New Record**.
    - (d) A button that **closes the form**, with the caption **Close**.
3. Arrange the buttons neatly in one row and make them the same size.
4. Switch to **Form View** and test each button. Take a screenshot of the form showing all four buttons.

Call the assessor to observe the buttons working.

Task 5: Change the Colours of the Form

Instructions:

1. Open `frmStudents` in **Design View**.
2. Change the **form background colour** to a light colour of your choice.
3. Change the **colour of the field labels** (text) to a dark colour that is easy to read.
4. Change the **background colour of each button** so that the **Close** button is a different colour from the other three buttons.
5. Change the **font colour** of the button captions so that the text is clearly visible.
6. Switch to **Form View**, save the form and take a screenshot of the finished form.

Task 6: Insert a Record Using the INSERT Command

Instructions:

1. Open a new query in **SQL View**.
2. Type an `INSERT INTO` command to add the record for **StudentID 106** shown in the second sample data table.
3. Run the query and save it as `qryInsert`.
4. Open the **Students** table and confirm that the new record appears. Take a screenshot.

Task 7: Retrieve Data Using SELECT query with a Condition

Instructions:

1. Open a new query in **SQL View**.
2. Type a `SELECT` command that displays **all fields** of students whose **Course is ICT**. Run it and save it as `qrySelectICT`. Take a screenshot of the result.
3. Type another `SELECT` command that displays only the **FirstName, LastName and Fees** of students whose **Age is greater than 20**. Run it and save it as `qrySelectAge`. Take a screenshot of the result.
4. In your answer sheet, write the number of records returned by each query.

Call the assessor to check the query results.

Task 8: Delete a Record Using the DELETE Command

Instructions:

1. Open a new query in **SQL View**.
2. Type a `DELETE` command that removes the record of the student with **StudentID 104** from the Students table.
3. Run the query and confirm the deletion when prompted. Save it as `qryDelete`.
4. Open the **Students** table and confirm that the record has been removed. Take a screenshot.
5. In your answer sheet, state why the `WHERE` condition is important in a `DELETE` command.

Call the assessor to check the table after deletion.

Task 9: Documentation and Saving

Instructions:

1. Check that your database contains the table `Students`, the form `frmStudents` and the queries `qryCreateTable`, `qryInsert`, `qrySelectICT`, `qrySelectAge` and `qryDelete`.
2. Save and close the database.
3. Submit your answer sheet containing the screenshots, the SQL commands you typed and your written answers.

THE END.